

Basics of computing

18/11/17
Saturday

Sum Rule - if an event can occur in m ways and another event can occur in n ways and if these events cannot occur simultaneously then one of the 2 events can occur in $m+n$ ways.

Product Rule - If an event H is decomposed into 2 diff. cases, then if an event A can occur in m ways and event B can occur in n diff ways then the task H can be performed in $m \times n$ ways.

Q. A book shelf has 6 different English books, 8 diff French books & 10 diff German books. Find how many ways of selecting:-
(i) 3 books from each language
(ii) 1 book in any one of the lang.

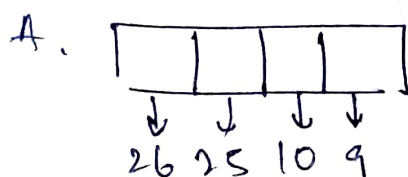
A. (i) $6 \times 8 \times 10 = 480$

(ii) $6 + 8 + 10 = 24$

Q. Suppose a person has 3 shirts and 5 ties, how many diff. ways of choosing a shirt and a tie?

A. $3 \times 5 = 15$

Q. Suppose we wish to construct a sequence of 4 symbols in which first 2 are English letters, next two are 8 digit nos. If no letter or digit can be repeated. Find the no. of diff sequence we can construct?



$$26 \times 25 \times 10 \times 9 = 58500$$

Q. Suppose a restaurant sells 6 South Indian dishes, 4 North Indian, 3 hot drink & 2 cold drinks for a breakfast. A student wishes to buy 1 SI dish and 1 hot drink or 1 NI dish & 1 cold drink.

A. $6 \times 3 + 4 \times 2$
 $= 18 + 8 = 26$

In how many ways can he buy his breakfast?

Q. There are 20 married couples in a party. Find the no. of ways of choosing 1 woman & 1 man from the party s.t. the 2 are not married to each other.

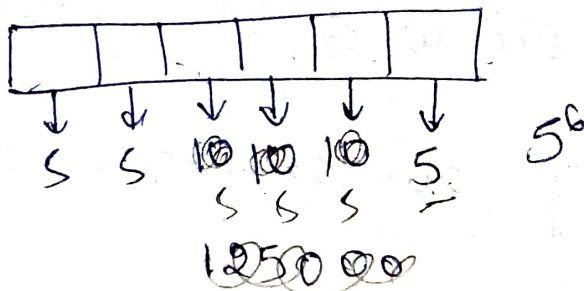
A. $20 \times 19 = 380$

Q. There are 4 bus routes b/w A and B and 3 bus routes b/w B & C. Find the no. of ways a person can make a round trip from A to A through B & C with repeating the ~~same~~ routes.

A. $4 \times 3 \times 4 \times 3 = 144$ (repetition allowed)

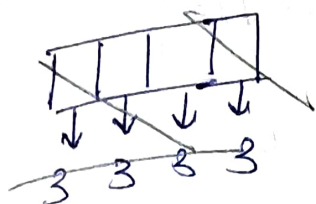


Q. A licence plate consists of 2 English letters which has only a, e, i, o, u followed by 4 digits if repetitions are allowed then how many licence plates will have a e i o u and even digits.



Q. Find the no of linear arrangement of 4 letters in BALL.

A.



81

if there are repeated letters, then.

$\frac{n!}{n_1! n_2! \dots}$ → no. of spaces

$$\frac{4!}{2!} = 12$$

no. of repeated letters

Q. No. of linear arrangement of 6 letters in PEPPER.

$$\frac{6!}{3! 2!} = \frac{6 \times 4 \times 2 \times 5}{2} = 12 \times 5 = 60$$

Q. Find the no. DATABASES

$$\frac{9!}{3! 2!} = 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 2 = 30240$$

Q. Find the no. of permutations of the letters of the word. MASSASSAUGA. In how many of these all 4 A's are together? How many of them begin with

$$(i) \frac{10!}{3! 4!} = 25200.$$

- consider the 4 A's as 1 block.

$$(ii) \frac{7!}{3!} = 840$$

$$(iii) \frac{9!}{3! 4!} = 7560$$

Q It is required to seat 5 men & 4 women in a row so that women occupy even places. How many such arrangements are possible?

A.

$$\textcircled{D} \quad \cancel{5 \times 4 \times 4 \times 3 \times 2 \times 1} \quad 5! \times 4! = 2880$$

$$5 \times 4 \times 4 \times 3 \times 3 \times 2 \times 2 \times 1 \times 1$$

Q. In how many ways can 6 men and 6 women be seated in a row

- (i) if any person may sit next to any other
(ii) if men and women must occupy alternate seats.

A. (i) $12! = 479,001,600$

(ii) $6! \cdot 6! \cdot 2 = 1,036,800$

Q. 4 diff maths books, 3 diff cs books, 2 diff english theory books are to be arranged in a shelf. How many arrangements are possible

- (i) if the books in each particular subject must be together? (ii) only math books must be together.

A. (i) $4! \times 3! \times 2! \times 3! \times 2!$ → to arrange the blocks.

(ii) $4! \times 8!$ 

Q Suppose we are interested in objects from a set of n , the set of r objects being selected is called combination of r — which is represented by ${}^nC_r = \frac{n!}{(n-r)!r!}$

Q An astronaut is having a dinner party for some members of her committee. Because of the size of her home, she can invite only 11 out of 20.

How many possibilities are there?

$${}^{20}C_4 = \frac{20!}{11!9!} = 167,960.$$

- Q. A gym teacher must select 9 girls from seniors and juniors if there are 25 seniors and 28 juniors.
- How many selections can be done?
 - If 2 juniors & 1 senior must be in a team
 - Team must have 4 juniors and 5 seniors.

A. (i) ${}^{53}C_9 = \underline{\underline{53C9}}$

(ii) ~~${}^{25}C_1 \times {}^{28}C_2$~~ $\cdot {}^{50}C_6 =$

(iii) ${}^{25}C_5 \times {}^{28}C_4$

Q. A gym teacher must make 4 volleyball teams of 9 girls each from 36 girls in how many ways can she select these 4 teams.

A. ${}^{36}C_9 \times {}^{27}C_9 \times {}^{18}C_9 \times {}^9C_9$

=

arrangement

Q. Find out all combinations of the word.

TALLAHASSEE and how many should not have adjacent A's.



A. (i) $11!$

$8! \cdot 2! \cdot 2! \cdot 2!$

T L L H S S E E
↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

(ii)

$\frac{9!}{2! \cdot 2! \cdot 2!}$

$\frac{8!}{2! \cdot 2! \cdot 2!} \times {}^9C_3$

Q A woman 11 close relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following ways:-

- (i) there are no restrictions on choices
- (ii) 2 particular persons will not attend separately
- (iii) 2 _____ together.

A. (i) ${}^{11}C_5$

(ii) ~~${}^{10}C_3$~~ ${}^9C_3 + {}^9C_5$

(iii) ~~${}^9C_5 + {}^{10}C_5$~~ $\underline{2 \times {}^9C_4 + {}^9C_5}$
 $\underline{378}$
 $\underline{378}$

Combination with repetition

21/11/17
Tuesday

Q. 3 scoops of ice cream from 5 flavours

b	c	d	s	v
---	---	---	---	---

$r+n-1$
 C_r

c c c $\rightarrow$ o o o $\rightarrow$ $\rightarrow$ $\rightarrow$
 b d v $\rightarrow$ o $\rightarrow$ $\rightarrow$ o $\rightarrow$ $\rightarrow$ o
 b v v $\rightarrow$ o $\rightarrow$ $\rightarrow$ $\rightarrow$ o o $\rightarrow$ o o

Q A donut shop offers 20 kinds of donuts assuming that there are atleast a dozen of each kind. In how many ways we can select a dozen of donuts?

$C_{12}^{20+12-1} = C_{12}^{31} = 141,120,525$

Q In how many ways we distribute 7 bananas & 6 oranges among 4 children so that each child receives atleast 1 banana.

$$C_{29}^{3+4-1} \times C_{06}^{6+4-1}$$

selecting r items
from n

$$20 \times 84 = 1680$$

A message is made up of 12 different symbols and is to be transmitted through a comm. channel. In add. to 12 symbols, the transmitter will also send 45 blank spaces b/w the symbols with atleast 3 spaces b/w each pair of consecutive symbols. In how many ways can the transmitter send such a msg?

$$A \quad 12! \times C_{12}^{11+12-1} = 12! \times C_{12}^{22} = 3.0974 \times 10^{14}$$

There are 12! ways to arrange 12 diff. symbols. and for each of these arrangements, there are 11 positions b/w the 12 symbols because there must be atleast 3 spaces b/w successive symbols, we use

up to 33 of 45 spaces and must now locate the remaining 12 spaces. This is now select with repetition of size 12 from a collection of size 11.

Q Consider the following prog. segment where i, j, k are the integer variables.

```

a for i = 1 to 20 do
  for j = 1 to i do
    for k = 1 to j do
      print (i + j + k);
    
```

How many times
the print statement
is executed?

$$\begin{matrix} i & 1 & 20 \\ j & 1 & i \\ k & 1 & j \end{matrix}$$

$$1 \times (2+1) \times ($$

selecti of 3 loops 20 times

$$1 \leq k \leq j \leq i \leq 20$$

$$C_3^{20+3-1} = C_3^{22}$$

$$= 1540$$

Q. Determine all integer solution of the equation
 $n_1 + n_2 + n_3 + n_4 = 7$, $n_i \geq 0$ ~~at~~ $1 \leq i \leq 4$

* Selecti of 7 items from a set of 4.

$$C_7^{4+7-1}$$

$$C_7$$

Q. In how many ways one can distribute 10 marbles amongst 6 different containers.

$$C_{10}^{10+6-1} = C_{10}^{15} = 3003$$

$$n_1 + n_2 + n_3 + n_4 + n_5 + n_6 = 10$$

Q. Determine the no. of integer solutions of
 $n_1 + n_2 + n_3 + n_4 = 32$

a) $n_i \geq 0$ $1 \leq i \leq 4$

b) $n_i > 0$ $1 \leq i \leq 4$

c) $n_1, n_2 \geq 5$ $n_3, n_4 \geq 7$

d) $n_i \geq 8$ $1 \leq i \leq 4$

A. a) ${}^{32+4-1}_{C_{32}}$

b) ${}^{32+4-1}_{C_{32}} - (4+4+3)$

c) $n=4$ $r=8$

${}^{28+4-1}_{C_{28}} = 4495$

$n_1 = 5, n_2 = 5, n_3 = 7, n_4 = 7$

$\Rightarrow$ these values already assigned, the rest is to be distributed.

d) 1

Q. In how many ways 10 marbles be distributed among 5 children.

- (i) no restriction
- (ii) if each child gets atleast 1 marble.

A. (i) ${}^{10+5-1}_{C_{10}} = 1001$

(ii) ${}^{5+5-1}_{C_5} = 126$

$(x+y)^n = \sum_{r=0}^n {}^nC_{n-r} x^{n-r} y^r$

Binomial Theorem

Q Find the binomial coeff of x^5 & y^2 in $(x+y)^7$

A. $(x+y)^7 = \sum_{r=0}^7 {}^7C_{7-r} x^{7-r} y^r$

for x^5 $7-r=5$
 $r=2$

${}^7C_5 = 14$

for y^2 $r=2$ ${}^7C_2 = 14$

Q Coeff of a^5 & b^2 in the exp. of $(2a-3b)^7$

A. $(2a-3b)^7 = \sum_{r=0}^n {}^nC_{n-r} (2a)^{n-r} (-3b)^r$

$$4-r=5$$

$$r=2$$

$${}^7C_5 \times 2^5 (-3)^2 = 14 \times$$

Q $(x+y)^4$

$$\sum_{j=0}^4 {}^4C_j x^{4-j} y^j$$

23/11/17
Thursday

Q $x^{12}y^{13}$ in $(x+y)^{25}$

$${}^{25}C_{12} = 5200300$$

Q. Explain multinomial theorem.

For the indices, n , the coeff. of $x_1^{n_1}, x_2^{n_2}, x_3^{n_3}, \dots$

in the expansion $(x_1 + x_2 + \dots + x_n)^n$ is

$$\frac{n!}{n_1! n_2! \dots n_t!} \text{ where each } n_i \text{ is an index with}$$

$$0 \leq n_i \leq n \quad \forall 1 \leq i \leq t \text{ and}$$

$$n_1 + n_2 + \dots + n_t = n$$

Q Find the coeff. of $x^2y^2z^3, xyz^5$ and x^3z^4 in the expansion of $(x+y+z)^7$

A. $x^2y^2z^3$ $\binom{n}{n_1 n_2 n_3} = \binom{7}{2 2 3} = \frac{7!}{2! 2! 3!} = 210$

$$xy^5 \quad \binom{7}{115} \quad \frac{7!}{5!1!1!} = 42$$

$$x^3z^4 \quad \binom{7}{304} = \frac{7!}{3!4!} = 35$$

Q. find the term which contains $x^{11}y^4$ in the exp of $(2x^3 - 3xy^2 + z^2)^6$

A. general term $\binom{6}{n_1 n_2 n_3} (2x^3)^{n_1} (-3xy^2)^{n_2} (z^2)^{n_3}$

$$\binom{6}{n_1 n_2 n_3} 2^{n_1} x^{3n_1+n_2} (-3)^{n_2} y^{2n_2} z^{2n_3}$$

$$3n_1 + n_2 = 11 \quad n_1 = 3$$

$$2n_2 = 4 \quad n_2 = 2$$

$$2n_3 = 0$$

$n_3 = 0$] no need to eliminate z can be given a power

$$\binom{6}{320} 2^3 x^{11} 9 y^4 z^0 = \frac{6!}{3!2!} \times 2^3 \times 9 x^{11} y^4$$

$$= \underline{\underline{4320 x^{11} y^4}}$$

Q. xyz^2 in $(2x - y - z)^4$

$$\binom{4}{n_1 n_2 n_3} (2x)^{n_1} (-y)^{n_2} (-z)^{n_3}$$

$$n_1 = 1 \quad n_2 = 1 \quad n_3 = 2$$

$$\frac{-4!}{1!1!2!} 2 x y z^2 = \underline{\underline{-24xyz^2}}$$

Q. $a^2 b^3 c^2 d^5 (a+2b-3c+2d+5)^{16}$

$$\binom{16}{n_1 n_2 n_3 n_4 n_5} = a^{n_1} (2b)^{n_2} (-3c)^{n_3} (2d)^{n_4} 5^{n_5}$$

$$n_1 = 2, n_2 = 3, n_3 = 2, n_4 = 5.$$

$$n_5 = 16 - 12 = 4$$

$$\frac{16!}{2! 3! 2! 5! 4!} 2^3 (-3)^2 2^5 5^4 a^2 b^3 c^2 d^5$$

$$= 403589 \times 10^{14} a^2 b^3 c^2 d^5$$

Principle of Inclusion and exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Consider a set S with cardinality of $S = n$ and the conditions C_i st. $1 \leq i \leq t$ satisfied by some elements of S . The no. of elements of set S that satisfy none of the conditions C_i is denoted by

$$\bar{N} = N(\bar{C}_1 \bar{C}_2 \bar{C}_3 \dots \bar{C}_t), \text{ given by the formula}$$

$$\bar{N} = N - [N(C_1) + N(C_2) + \dots + N(C_t)] +$$

$$- 2[N(C_1 C_2) + N(C_1 C_3) + \dots + N(C_1 C_t) + \dots]$$

$$+ N(C_{t-1} C_t)] - [N(C_1 C_2 C_3) + N(C_1 C_2 C_4) + \dots + N(C_{t-2} C_{t-1} C_t)] + \dots (-1)^t N(C_1 C_2 \dots C_t)$$

OR

$$N = N - \sum_{1 \leq i \leq t} N(C_i) + \sum_{1 \leq i \leq j \leq t} N(C_i C_j) + \dots$$

$$\dots + \sum_{1 \leq i \leq j \leq k \leq t} N(C_i C_j C_k) + \dots + (-1)^{t+1} N(C_1 C_2 \dots C_t)$$

$$\begin{aligned} N(C_1) &= |A| \\ N(C_2) &= |B| \\ N(C_1 C_2) &= |A \cap B| \\ |\bar{A} \cap \bar{B}| &= N - |A \cup B| \end{aligned}$$

Q. Among the students in a hostel 12 students study maths, 20 study chem & 8 study biology, 20 physics

$$\begin{aligned} |A| &= 12 & |A \cap B| &= 5 & |B \cap D| &= 4 \\ |B| &= 20 & |A \cap C| &= 7 & |C \cap D| &= 3 \\ |C| &= 20 & |A \cap D| &= 4 & |A \cap B \cap C| &= 3 \\ |D| &= 8 & |B \cap C| &= 16 & |A \cap B \cap D| &= 2 \end{aligned}$$

$$|\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| = 71$$

$$|B \cap C \cap D| = 2, |A \cap B \cap D \cap C| = 2$$

$$|A \cap C \cap D| = 3 \quad \text{Find total no. of student}$$

$$\begin{aligned} \underline{A} \quad N &= |\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}| + |A \cup B \cup C \cup D| \\ &= 71 + 12 + 20 + 20 + 8 - 5 - 7 - 4 - 16 - 4 - 3 \\ &\quad + 3 + 2 + 2 + 3 - 2 \\ &= 100 \end{aligned}$$

[Signature]

Q. How many integers between 6001 & 300

(i) divisible by 6 at least one of

(a) 5, 6, 8 (ii) none of 5, 6, 8.

37

$$A \quad |A \cup B \cup C| = S_1 - S_2 + S_3 \quad \sum N(C_i C_j C_k)$$

$\downarrow \quad \downarrow$
 $\sum N(C_i) \quad \sum N(C_i C_j)$

$$S_1 = N(C_1) + N(C_2) + N(C_3)$$

$$= 60 + 50 + 37$$

$$S_2 = N(C_1 C_2) + N(C_1 C_3) + N(C_2 C_3)$$

$$= 10 + 12 + 7$$

$$S_3 = N(C_1 C_2 C_3) = 2$$

$$N = 120$$

$$(ii) \quad 300 - 120 = 180$$

Q. Determine the no. of +ve integers b/w $1 \leq i \leq 100$ which are not divisible by 2, 3 or 5.

27/11/17
Monday

A. 26

$$N(C_1) = 50$$

$$N(C_2) = 33$$

$$N(C_3) = 20$$

$$\overline{N} = 100 - (50 + 33 + 20 - 16 - 6 - 10 + 3)$$

$$N(C_1 C_2) = 16 \quad N(C_1 C_3) = 10$$

$$N(C_2 C_3) = 6$$

$$N(C_1 C_2 C_3) = 3$$

$$N(C_1 C_2 C_3) = 3$$

Q. In how many ways can 26 letters of an alphabet be permuted so that none of the patterns car, dog, pun or hile occurs

$$N = 26!$$

A. $N(C_1) = (23+1)!$ considering car as 1 letter & removing it from the 26 letters

$$N(C_2) = 24!$$

$$N(C_3) = 24!$$

$$N(C_4) = 23!$$

$$N(C_1 C_2) = (26 - 6 + 2)! = 22!$$

$$N(C_2 C_3) = 22!$$

$$N(C_3 C_4) = 21!$$

$$N(C_1 C_3) = 22!$$

$$N(C_1 C_4) = 21!$$

$$N(C_2 C_4) = 21!$$

$$N(C_1 C_2 C_3) = 20! \quad 26 - 9 = 17 + 3$$

$$N(C_2 C_3 C_4) = 19!$$

$$N(C_1 C_2 C_4) = 19!$$

$$N(C_1 C_2 C_3 C_4) = 17!$$

$$\bar{N} = N - |A \cup B \cup C \cup D|$$

$$= N - [24! \times 3 + 23! - (3 \times 22! + 3 \times 21!)$$

$$+ 19! \times 2 + 20! - 17!]$$

$$= 4.014 \times 10^{26}$$

Q. Find the no. of permutations from a word from which none of the patterns spin, game, path or net occurs.

spinel

A. $N = 26!$

$$N(C_1) = 23!$$

$$N(C_2) = 23!$$

$$N(C_3) = 23!$$

$$N(C_4) = 24!$$

$$N(C_1 C_2) = 20!$$

$$N(C_1 C_3) = 0 \Rightarrow \text{when letters are repeated}$$

$$N(C_1 C_4) = 0$$

$$N(C_2 C_3) = 0$$

$$N(C_2 C_4) = 0$$

$$N(C_3 C_4) = 0$$

$$N(C_1 C_2 C_3) = 0$$

$$N(C_2 C_3 C_4) = 0$$

$$N(C_1 C_3 C_4) = 0$$

$$N(C_1 C_2 C_3 C_4) = 0$$

$$\textcircled{a} \quad \bar{N} = 26! - (23! \times 3 + 24! - 20!)$$

$$= 4.0259 \times 10^{26}$$

Q. Determine no. of permutations of the letters

T N U I S G R E A T such that none of the words

T N U, I S, G R E A T occur as consecutive letters

such as T S T N U G R E A T, U N T G R E A T S,

T N U T U I S G R E A

etc. are not allowed

$$N = 10!$$

$$N(C_1) = 8!$$

$$N(C_2) = 9!$$

$$N(C_3) = 6!$$

$$N(C_1 C_2) = 7!$$

$$N(C_2 C_3) = 5!$$

$$N(C_1 C_3) = 4!$$

$$N(C_4 C_2 C_3)$$

$$= 3!$$

$$\bar{N} = 10! - (8! + 9! + 6! - 7! - 5! - 4! + 3!).$$

$$= 3230058$$

Find the no. of non-negative integer sol. of 6 eq.
 $x_1 + x_2 + x_3 + x_4 = 18$ s.t. $x_i \leq 7$ for $1 \leq i \leq 4$.

$$A. \quad N = C_{18}^{18+4-1}$$

~~(18+4-1)C_{18}~~

$$N(C_1) = x_i \geq 8$$

$$= {}^{10}C_{(18-8)}$$

$${}^{10+4-1}C_{10}$$

$$N(C_2) = 10$$

$$N(C_3) = 10$$

$$N(C_4) = 10 \quad {}^{13}C_{10}$$

$$N(C_1 C_2) = N(C_2 C_3) = N(C_3 C_4) = N(C_1 C_3) = N(C_2 C_4) = N(C_1 C_4) = 5C_2$$

$$N(C_1 C_2 C_3) = N(C_2 C_3 C_4) = N(C_1 C_3 C_4) = 0$$

$$N(C_1 C_2 C_3 C_4) = 0$$

$$\bar{N} = {}^{21}C_{18} - (4 \times {}^{13}C_{10} - 6 \times {}^5C_2)$$

$$= 246$$

$\therefore 246$ solutions are there for the given condition

Q. Determine how many integer solutions are there
 $n_1 + n_2 + n_3 + n_4 = 19$

a) $0 \leq n_i$ for all $1 \leq i \leq 4$

b) $0 \leq n_i \leq 8$ for all $1 \leq i \leq 4$

A. a) $\frac{19+4-1}{C_{19}} = 1540$

b) $N = 1540$

$N(C_1) = \overset{13}{C_{10}} = N(C_2) = N(C_3) = N(C_4)$

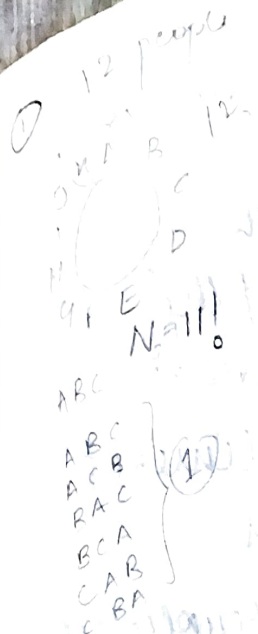
$N(C_1 C_2) = N(C_2 C_3) = N(C_3 C_4) = N(C_1 C_3) = N(C_1 C_4)$
 $= N(C_2 C_4) = 4C_1$

$N(C_1 C_2 C_3) = N(C_2 C_3 C_4) = N(C_1 C_3 C_4) = 0$
 $N(C_1 C_2 C_3 C_4) = 0$

$\bar{N} = 1540 - [4 \times 13C_{10} - 6 \times 4C_1]$
 $= \underline{\underline{372}}$

Q. ① 6 married couples are to be seated at a circular table. In how many ways can they arrange themselves so that no wife sits next to her husband.

Q. ② In certain areas of the countryside there are 5 villages. An engg. is to devise a system of 2 way roads so that after the system is completed no village will be isolated. In how many ways can he do this?



$$N(1) = \text{from couple is sealed together} = 10!$$

$$N(C_1) = 10! - N(C_2) - N(C_3) - N(C_4) - N(C_5) = N(6)$$

$$S_1 = 6 \times 10!$$

$$N(C_2) = 9! \quad S_2 = 15 \times 9!$$

$$N(C_1 C_2 C_3) = 8! \quad S_3 = 20 \times 8!$$

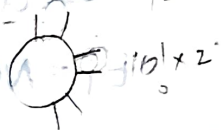
$$N(C_1 C_2 C_3 C_4) = 7! \quad S_4 = 15 \times 7!$$

$$N(C_1 C_2 C_3 C_4 C_5) = 6! \quad S_5 = 6 \times 6!$$

$$N(C_1 C_2 C_3 C_4 C_5 C_6) = 5! \quad S_6 = 5!$$

$$\bar{N} = 11! - [2 \times 6 \times 10! - 2 \times 15 \times 9! + 2 \times 20 \times 8! - 2 \times 15 \times 7! + 2 \times 6 \times 6! - 2 \times 5 \times 5!]$$

$$= 2282220$$



② (A)

$$N(C_1) = 4C_2 + 4C_3 + 4C_4$$

$$N(C_2) = 4C_2 \quad S_1 = 5 \times 4C_2$$

$$N(C_3) = 4C_2 + 4C_3$$

$$N(C_4) = 4C_2 + 4C_3 + 4C_4$$

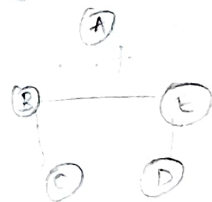
$$N(C_5) = 4C_2 + 4C_3 + 4C_4 + 4C_5$$

$$N(C_1 C_2) = 3C_2 \quad S_2 = 10 \times (3C_2 + 3C_3)$$

$$N(C_1 C_2 C_3) = 2C_2 \quad S_3 = 10$$

$$N(C_1 C_2 C_3 C_4) = 0 \quad S_4 = 0$$

$$N(C_1 C_2 C_3 C_4 C_5) = 0 \quad S_5 = 0$$



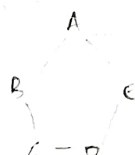
$$\bar{N} = 5C_2 - [5 \times 4C_2 - 10 \times 3C_2 + 10]$$

$$\bar{N} = N - [S_1 + S_2 + S_3 + S_4 + S_5]$$

$$N = 5C_2 + 5C_3 + 5C_4 + 5C_5 = 26$$

$$1.26$$

$$\bar{N} = 26$$



Generalization of principal

28/11/17
Tuesday

keywords: exactly, atleast

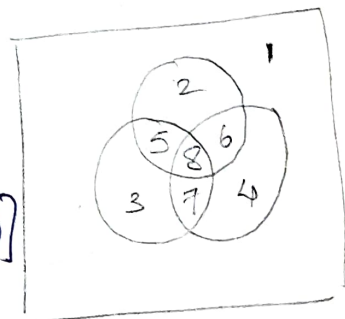
Consider a set S with cardinality of $|S|=N$ with conditions $C_1, C_2 \dots C_t$ satisfied by some of the elements of S . The inclusion & exclusion provides a way to determine $N(\bar{C}_1 \bar{C}_2 \bar{C}_t)$ i.e. no. of elements in S that satisfies none of the t conditions

Now we want to determine E_m which denotes no. of elements in S that satisfies exactly m of t conditions

$$E_1 = N(C_1 \bar{C}_2 \bar{C}_3 \dots \bar{C}_t) + N(\bar{C}_1 C_2 \bar{C}_3 \dots \bar{C}_t) \\ + \dots + N(\bar{C}_1 \bar{C}_2 \dots \bar{C}_{t-1} C_t)$$

$$E_2 = N(C_1 C_2 \bar{C}_3 \dots \bar{C}_t) + N(\bar{C}_1 C_2 C_3 \bar{C}_4 \dots \bar{C}_t) \\ + \dots + N(\bar{C}_1 \bar{C}_2 \dots C_{t-1} C_t)$$

$$E_1 = N(C_1) + N(C_2) + N(C_3) \\ - 2[N(C_1 C_2) + N(C_2 C_3) + N(C_1 C_3)] \\ + 3N(C_1 C_2 C_3)$$



$$\Rightarrow \boxed{E_1 = S_1 - 2S_2 + 3S_3}$$

$$\boxed{E_2 = S_2 - 3S_3}$$

} for $t=3$

5x
times

count elements in
region 5, 6, 7
twice &
and those in
region 8
thrice

$$E_m = S_m - \binom{m+1}{1} S_{m+1} + \binom{m+2}{2} S_{m+2} - \dots + (-1)^{t-m} \binom{t}{t-m} S_t$$

for $t=4$,

$$E_1 = E_1 - \binom{2}{1} S_2 + \binom{3}{2} S_3 - \binom{4}{3} S_4$$

$$L_m = S_m - \binom{m}{m-1} S_{m+1} + \binom{m+1}{m-1} S_{m+2} - \dots + (-1)^{t-m} \binom{t+1}{m-1} S_t$$

(last) is at least 2 conditions

Q. In how many ways can the letters in the word ARRANGEMENT be arrayed so that

- (i) there are exactly 2 pairs of consecutive identical letters
- (ii) all are ~~different~~ letters

A.U) $N = 11!$
 $\underline{2! \cdot 2! \cdot 2! \cdot 2!}$

$$N(C_1) = 2 \text{ consecutive A's} = \frac{10!}{(2!)^3}$$

$$N(C_2) = R's = \frac{10!}{(2!)^3}$$

$$N(C_3) = E's = \frac{10!}{(2!)^3}$$

$$N(C_4) = N's = \frac{10!}{(2!)^3}$$

$$S_1 = \frac{10!}{(2!)^3} \times 4$$

$$N(C_1 C_2) = \frac{9!}{(2!)^2}$$

$$N(C_1 C_3) =$$

$$S_2 = \frac{6 \times 9!}{(2!)^2}$$

$$N(C_1 C_4) =$$

$$N(C_2 C_3) =$$

$$N(C_2 C_4) =$$

$$N(C_3 C_4) =$$

$$N(C_1 C_2 C_3) = \frac{8!}{2!}$$

$$N(C_1 C_2 C_4) =$$

$$S_3 = \frac{3 \times 8!}{2!}$$

$$N(C_2 C_3 C_4) =$$

$$N(C_1 C_2 C_3 C_4) = 7! = S_4$$

$$\begin{aligned} \text{(i)} E_2 &= S_2 - \binom{3}{1} S_3 + \binom{4}{2} S_4 \\ &= \frac{544320}{20160} - 3 \times 60480 + 6 \times 5040 \\ &= \frac{190980}{332640} \end{aligned}$$

$$\begin{aligned} \text{(ii)} L_m &= S_2 - \binom{2}{1} S_3 + \binom{3}{2} S_4 \\ &= \frac{544320}{272160} - 2 \times 60480 + 3 \times 5040 \\ &= \frac{116320}{398160} \end{aligned}$$



Q. In how many ways can one arrange the letters in the word C O R R E S P O N D E N T S so that-

- (i) there is no pair of consecutive identical letters
 (ii) there are exactly 2 pair _____
 (iii) at least 3 pair of consecutive _____

A. $N = \frac{14!}{2!2!2!2!2!}$

$$\left. \begin{array}{l} N(C_1) \\ N(C_2) \\ N(C_3) \\ N(C_4) \\ N(C_5) \end{array} \right\} = \frac{13! \times 5}{(2!)^4}$$

$$N(C_2 C_4 C_5)$$

$$N(C_1 C_4 C_5)$$

$$N(C_1 C_2 C_3)$$

$$N(C_1 C_2 C_4)$$

$$N(C_1 C_2 C_5)$$

$$N(C_2 C_3 C_4)$$

$$N(C_2 C_3 C_5)$$

$$N(C_3 C_4 C_5)$$

$$\frac{5!}{3!2!}$$

$$S_3 = \frac{11! \times 10}{(2!)^2}$$

$$N(C_1 C_2)$$

$$N(C_1 C_3)$$

$$N(C_1 C_4)$$

$$N(C_1 C_5)$$

$$N(C_2 C_3)$$

$$N(C_2 C_4)$$

$$N(C_2 C_5)$$

$$N(C_3 C_4)$$

$$N(C_3 C_5)$$

$$N(C_4 C_5)$$

$$N(C_1 C_2 C_3 C_4)$$

$$N(C_1 C_2 C_3 C_5)$$

$$N(C_2 C_3 C_4 C_5)$$

$$N(C_1 C_3 C_4 C_5)$$

$$N(C_1 C_2 C_4 C_5)$$

$$N(C_1 C_2 C_3 C_4 C_5) = 9!$$

$$\frac{5 \times 10!}{(2!)^2}$$

$$(ii) \overline{N} = N - [S_1 - S_2 + S_3 - S_4 + S_5]$$

$$= \frac{14!}{(2!)^5} - \left[\frac{5 \times 13!}{(2!)^4} - \frac{10 \times 12!}{(2!)^3} + \frac{10 \times 11!}{(2!)^2} \right]$$

$$= 1286046720$$

$$- \frac{5 \times 10! + 9!}{(2!)} \Bigg\}$$

$$(ii) E_2 = S_2 - \binom{3}{1} S_3 + \binom{4}{2} S_4 - \binom{5}{3} S_5$$

$$= \cancel{5 \times 10!} - \frac{5 \times 13!}{16} - \frac{3 \times 10 \times 12!}{8} + \frac{6 \times 10! \times 11!}{4}$$

$$+ 10!$$

$$= 350,179,200$$

$$(iii) L_3 = S_3 - \binom{3}{2} S_4 + \binom{4}{2} S_5$$

$$= 74,753,280$$

Pigeonhole Principle

Q. In how many ways can one arrange all of the letters in the word information so that no pair of consecutive letters occurs more than once.
 (Here we want to count the arrangement such as (IINNOOFRTMA, FORTMAII, NON) but not (INFORINMOTA (occure IN. occure twice)
 NORTFNIOIAM (---NO---))

Ans

N
O
I

N N
O O

Pigeon hole Principle

If m pigeons occupy n pigeonholes and $m > n$, then at least 1 pigeonhole must contain 2 or more pigeons in it.

General

If m pigeons occupy n ph then at least 1 pigeonhole must contain $p+1$ or more pigeons where $p = \frac{m-1}{n}$

Q. A bag contains 12 pairs of socks (each pair in diff. colour). If a person draws the socks 1 by 1 at random, determine at most how many draws are required to get at least 1 pair of matching socks.

A. 13

Q. Prove that in any set of 29 persons at least 5 persons must have been born on the same day of the week

A. no. of days = 7
~~evening~~

Perangements Derangement

30/11/17
Thursday

Q A permuatāli of n distinct objects in which none of the objects is in its natural place is called derangement.

Q For eg. permuatāli of integers of 1 to n where 1 is not in 1st place, 2 is not in 2nd place & so on. find the no. of derangements of 1, 2, 3, 4.

$$d_4 = 4! \left\{ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right\}$$

$$= 24 \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right)$$

$$= 12 - 4 + 1 = \underline{\underline{9}}$$

(list it)

2413

3421

$$d_n = n! \left\{ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right\}$$

$$= n! \times \sum_{k=0}^n \frac{(-1)^k}{k!}$$

Q. Find d_5, d_6, d_7 & d_8 .

A. $d_5 = 5! \left\{ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right\} = 44$

$$d_6 = 265$$

$$d_7 = 1854$$

$$d_8 = 14833.$$

$\frac{1}{3} V_{cc}$

o. vol - 1 0

$\frac{2}{3} V_{cc}$

Rook Polynomial

Q.

1	2	3
4		
	5	6

formula:

$$r(c, x) = 1 + r_1 x + r_2 x^2 + r_3 x^3 + \dots$$

$$r(c, x) = 1 + 6x + 8x^2 + 2x^3$$

r_0

$$\{1, 5\} \quad \{2, 4\} \quad \{3, 4\} \quad \{4, 2\} \quad \{4, 5\}$$

$$\{1, 6\} \quad \{2, 6\} \quad \{3, 5\} \quad \{4, 5\} \quad \{4, 6\}$$

r_3

$$\{1, 5, 6\} \quad \{3, 4, 5\} \quad \{2, 4, 6\}$$

Q.

1	2
3	4

$$r(c, x) = 1 + 4x + 2x^2$$

non latin rooks = not present in same row or column

$$r_0 = 1 \text{ (always)}$$

$$r_1 = 1 \text{ non latin rooks}$$

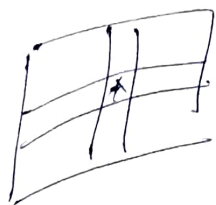
$$r_2 = 2$$

$$r_3 = 3$$

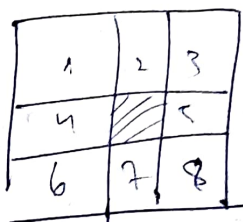
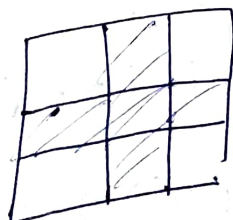
Q. Find the rook polynomial of 3×3 chessboard.

Expansion formula. $r(C, n) = n r(D, n) + r(E, n)$.

make a stair as you wish
D board is made by removing last
row & column.



removing the
last row
with
stair



D

E

$$r(E, n) = 1 + 8n + 14n^2 + \cancel{6}n^3$$

$$r(D, n) = 1 + 4n + 2n^2$$

$$r(C, n) = 1 + 8n + 14n^2 + \cancel{6}n^3 + n + 4n^2 + 2n^3$$

$$= 1 + 9n + 18n^2 + \cancel{7}n^3$$

(1,2) (1,4) (1,5) (1,7) (1,8)

(2,4) (2,6) (2,7) (4,8)

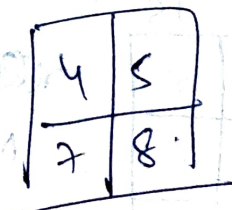
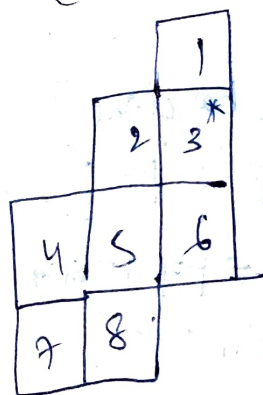
(5,7) (6,7) (6,8)

(7,8)

(1,5,7), (2,4,8)

(3,4,7) (6,2,5)

(8,4,2)



D

$$r(C, n) = 1 + 4n + 2n^2$$

$$r(C, n)$$

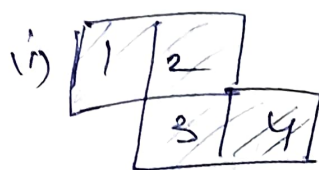
$$= 1 + 7n + 12n^2 + \cancel{3}n^3$$

(1,2,4) (1,4,8)

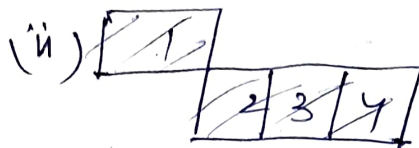
(1,2,7) (2,6,7) (1,5,7) (1,6,7) (1,6,8)

$$r(C, n) = 1 + 8n + 16n^2 + \cancel{9}n^3$$

Q. Find the rook polynomial for the shaded chess boards C_1 & C_2 .



C_1



C_2

$$r(C_1, n) = 1 + 4n + 3n^2$$

$$r(C_2, n) = 1 + 4n + 6n^2 + 3n^3$$

Q. In making seating arrangement for their son's wedding reception Grace & Nick are down to 4 relatives $R_i, 1 \leq i \leq 4$ who do not get along with one another. There is a single open seat at each of the 5 tables $T_j, 1 \leq j \leq 5$. Because of the family difference

- R_1 will not sit at T_1 or T_2
- R_2 _____ T_2
- R_3 _____ T_3 or T_4
- R_4 _____ T_4 or T_5

4p3

How many no. of arrangements can be done?

using principle of inclusion & exclusion

A.

	T_1	T_2	T_3	T_4	T_5
R_1	X	X	1	2	3
R_2	4	X	5	6	7
R_3	8	9	X	X	10
R_4	11	12	13	X	X

$$N(C_1) = 2 \times 4!$$

$$N = SP_4 = 5!$$

$$N(C_2) = 4!$$

$$N(C_3) = 2 \times 4!$$

$$N(C_4) = 2 \times 4!$$

$$N(C_1 C_2) = 3!$$

$$N(C_1 C_3) = 4 \times 3!$$

ii) R_1 at T_1 R_3 at T_3

(ii) R_1 at T_2 R_3 at T_3

(iii)

(iv)

$$N(C_3 C_4) = 3 \times 3!$$

$$N(C_2 C_4) = 3 \times 3!$$

$$N(C_2 C_3) = 3 \times 3!$$

$$N(C_1 C_2 C_3) = 2 \times 2!$$

$$\underline{A.} = 25.$$

$$1 + 7n + 16n^2 + 13n^3 + 3n^4$$